

AI-Supported Digital Health Navigation and Conversational Analytics for Refugee Health Service Utilization

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Hypothesis (open - reconcile with Aral + Shatha before final PDF): Hybrid Fork B + Fork C: test whether AI-mediated digital health tools increase utilization of general and public health services among refugee and returnee populations, using conversational health data as a novel implementation-science evidence modality versus survey-only workflows.

1. Transformative Potential and Impact

Humanitarian health programs still rely heavily on structured survey instruments to understand needs, barriers, and health-seeking behavior in displacement settings. These tools capture predefined fields but miss conversational nuance: how people describe symptoms, express mistrust, navigate entitlement complexity, or decide when to seek facility-based care. At the same time, AI-assisted digital health tools are spreading in stable health systems, yet rigorous evidence on their use in humanitarian response remains thin, especially for mobile refugee and returnee populations who cannot be served by fixed-facility models alone.

This project proposes a bounded, high-risk catalytic study at the intersection of public health, implementation science, and data science: whether AI-supported digital health navigation, paired with systematic analysis of de-identified conversational health data, can increase utilization of general and public health services among refugee and returnee populations in resource-limited settings. The scientific contribution is twofold. First, it treats operational conversational datasets as a new evidence modality for humanitarian maternal and reproductive health, complementing rather than replacing survey-based workflows. Second, it applies rapid-cycle implementation learning appropriate to humanitarian timelines, rather than defaulting to slow, large-scale trials ill-suited to a two-year Individual Award.

The Center for Humanitarian Health (CHH) at Johns Hopkins is positioned to lead this evidence generation. HERA Digital Health, a humanitarian digital health organization with operational reach among displaced populations, provides a bounded data partnership (in-kind access and technical support within the $\leq 10\%$ subaward cap). This award does not fund large-scale field implementation; it seeds the science and evaluation architecture that could support follow-on external funding.

Translational impact: Better targeting of digital navigation interventions; actionable intelligence on barriers to public-health service use; trainee pipeline through a DrPH dissertation thread; publication pathway on AI in humanitarian health evidence.

2. Strategic Alignment

This proposal aligns directly with the Life Sciences Research Initiative mandate for bold, high-risk, high-reward science across life sciences, including public health, medicine, and health policy. It explicitly integrates data science and AI (NLP, pattern detection, compute infrastructure) with population-health translation.

Strategically, the project advances CHH's role as a center that can generate rigorous evidence on

emerging technology in humanitarian response, without duplicating the separate \$12M anticipatory-action team expression of interest (Shatha Elnakib serves as co-PI on that distinct project). The Individual Award is a separate, faculty-led pilot focused on digital health utilization and conversational analytics in displacement settings.

The project meets Shatha-team design gates: AI is central to methods and intervention framing; life-science relevance is carried through maternal and reproductive health access among displaced populations; student investment is explicit through DrPH trainee support.

3. Strength of Approach and Feasibility

Scientific question: Among refugee and returnee populations in settings with limited resources or strained health infrastructure, does exposure to AI-supported digital health navigation improve utilization of general and public health services relative to pre-intervention or comparison benchmarks, and what implementation factors explain uptake?

Design: Mixed-methods implementation science over 24 months. We do not propose a large RCT; the program does not require one. Instead:

Data sources: De-identified conversational health data from HERA humanitarian operations (latent asset; reduces field startup). Where feasible, comparison to survey-based benchmarks (e.g., structured data collection workflows). Geography: Initial analytic focus on refugee/returnee corridors where CHH and partners have access; specific site selection will follow IRB and partner agreement review before naming in the final submission.

AI / data science methods:

- NLP and thematic coding of health-seeking narratives
- Pattern detection for barriers (entitlement confusion, facility access, trust)
- Bias and safety review of AI-mediated outputs
- JHU-hosted compute (allowable per RFP)

Ethics: All conversational data handled under JHU IRB with explicit de-identification, no PII/PHI in shared analytic artifacts, and protection-aware protocols for vulnerable populations.

Feasibility: JHU hosts analysis, faculty effort, trainee support, and compute. HERA provides data access and technical partnership within $\leq 10\%$ subaward or in-kind. Prior quasi-experimental work on antenatal visit attendance in Istanbul migrant health centers (JMIR preprint under peer review; NCT05094518) demonstrates CHH-adjacent evaluation capacity, though that study used smartphone reminders rather than conversational AI and should not be over-claimed here.

4. Team Capability and Environment

Environment: JHU CHH provides humanitarian health evaluation rigor; CBID provides engineering and AI institute connectivity; HERA provides operational credibility without JHU hosting large-scale implementation. Small roster by design (no extra maternal faculty) per team discipline.

Multidisciplinarity: Public health (CHH) + bioengineering/AI (CBID) + implementation partner capacity (HERA, capped).

5. Future Funding Potential

This Individual Award is designed as a springboard, not a terminal grant.

HERA field implementation at scale would be funded externally after this award establishes the evidence base. Institutional funds here buy catalytic science, IRB-ready methods, and utilization metrics that make external proposals competitive.

6. Appropriate Use of Institutional Funds and Budget Justification

Requested amount: \$300,000 over 2 years (verify with Shatha/JHURA; range \$200k-\$500k allowed).

Budget discipline: No large external subs; JHU-hosted salaries and compute dominate; HERA line stays at or below 10%. Service contract vs subaward distinction: verify with JHURA.

7. Demonstrated Need for Support

This project is not yet competitive for traditional external mechanisms at full implementation scale. The Individual Award provides catalytic institutional investment to:

- Establish IRB-ready conversational analytics methods in humanitarian settings
- Generate utilization-focused implementation evidence
- Position the faculty team for DIV/NIH/foundation follow-on

If prior DIV or related external reviews are attachable and scored highly without funding, they may supplement (not replace) this internal case.

8. Data Science and AI Relevance

AI is not peripheral; it is the analytic engine:

- NLP on conversational health text in displacement contexts
- LLM-mediated navigation tools evaluated for utilization impact
- Pattern detection for health-seeking behavior and system barriers
- Safety/bias monitoring for AI outputs in vulnerable populations
- Compute budget line explicitly requested

References (beyond 4-page limit): Add after pagination check.

Red-team pass applied: Institutional science-gap open; HERA capped at 10%; Aral trainee-only; Yazdi framed as proposed collaborator pending scoping; removed unverified corpus volume; geography softened pending IRB; separated \$12M Eol; no em dashes; JMIR preprint scoped correctly.

Answer entry used: none - full narrative PDF, not gated answer-library field